

Deletion Bridges & Final Rearrangement Structure

**What Deletion Bridges Signify In Terms of Baca's Chains and
What The Final Rearranged Structure of Baca Chain Looks Like.**

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An Overview on Patient P05-1657

This is Baca's Chain for patient P05-1657

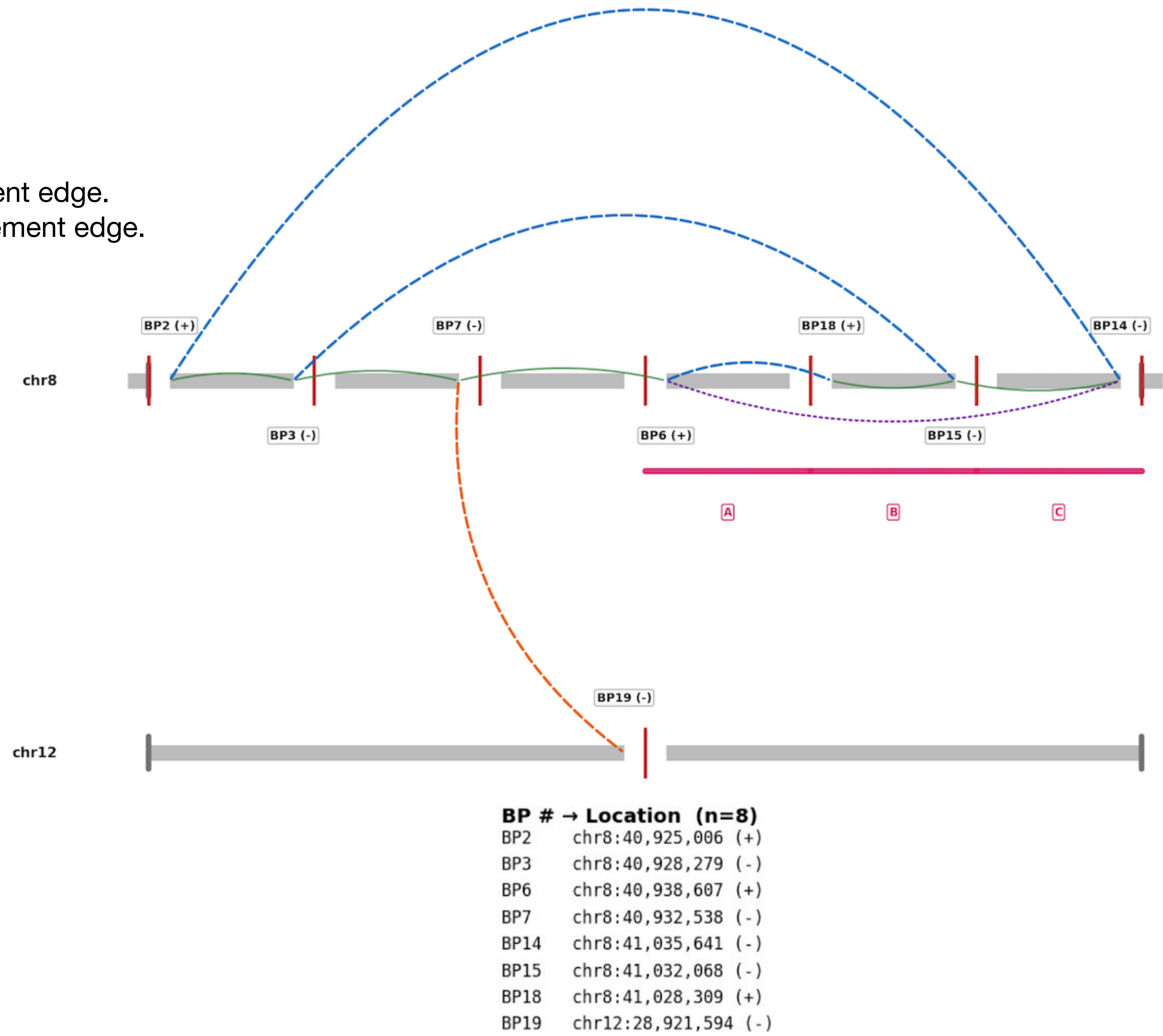
Here,
The **Blue Dashed-Edges** show the **INTRA-Chromosomal** Rearrangement edge.
The **Orange Dashed-Edges** show the **INTER-Chromosomal** Rearrangement edge.
The **Purple Circled-Edges** show the **Deletion Bridge**.
The **Green** Edges show the **Adjacency Edge**.

An **Adjacency Edge** simply means that the breakpoints are “Physically” next to each other. Their genomic coordinates also confirm this.

In our chain, the breakpoints are already arranged in increasing order of location. The adjacency edges just confirm that a certain set of breakpoints are next to each other, not all.

For example, BP6 and BP18 are INFERRED to be next to each other rather than confirmed. Baca confirmed adjacent breakpoints using Pxy.

“calculated the probability that any pair of neighboring DNA breakpoints *X* and *Y* would arise independently of each other (*PXY*) on the basis of (1) their reference genome distance and (2) the local rate of rearrangements observed in our tumor panel”
~ **Baca S C et al 2013 Punctuated evolution of prostate cancer genomes**



What Are Deletion Bridges Exactly?

What Baca means by Deletion Bridges

A **deletion bridge** is Baca's name for a stretch of DNA that looks deleted, where the two ends of that missing stretch each belong to a different rearrangement rather than to each other.

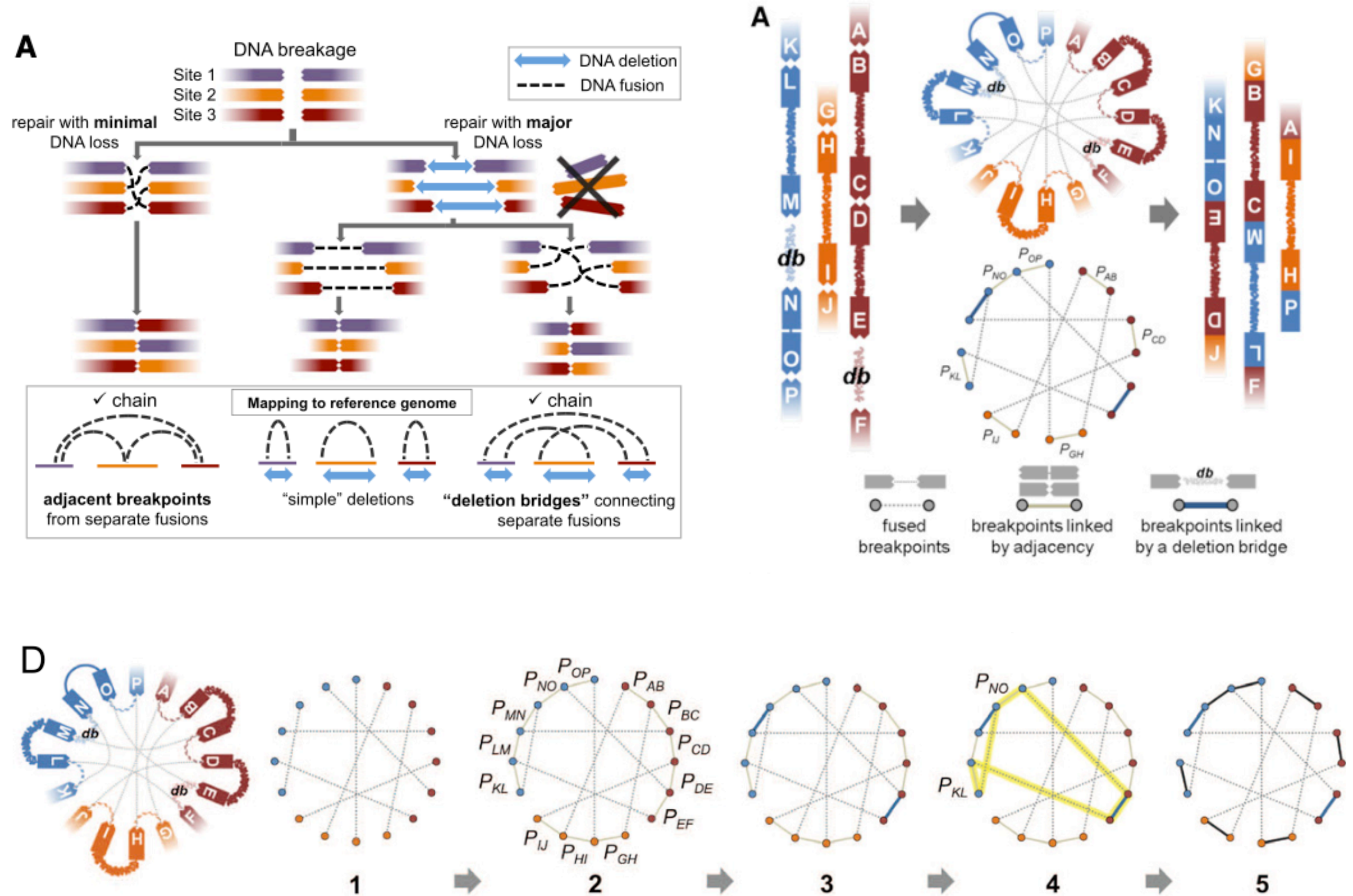
Baca confirms it's a **real deletion** in two ways:

1. The copy-number data shows an **actual drop** across **that stretch**.
2. The sequencing reads at both ends point into the **missing region**, consistent with a **real break** there rather than noise.

A deletion bridge connects two **separate rearrangements** and **ChainFinder** treats it as one of three ways breakpoints can be linked together into the same coordinated “**chain**.”

Baca treats this as real, **meaningful loss of DNA** — **distinct** from the **minimal loss** expected from **ordinary repair**

He also mentions specific tumor-suppressor genes (CDKN1B, ETV6, ETV3) as having been lost this way.



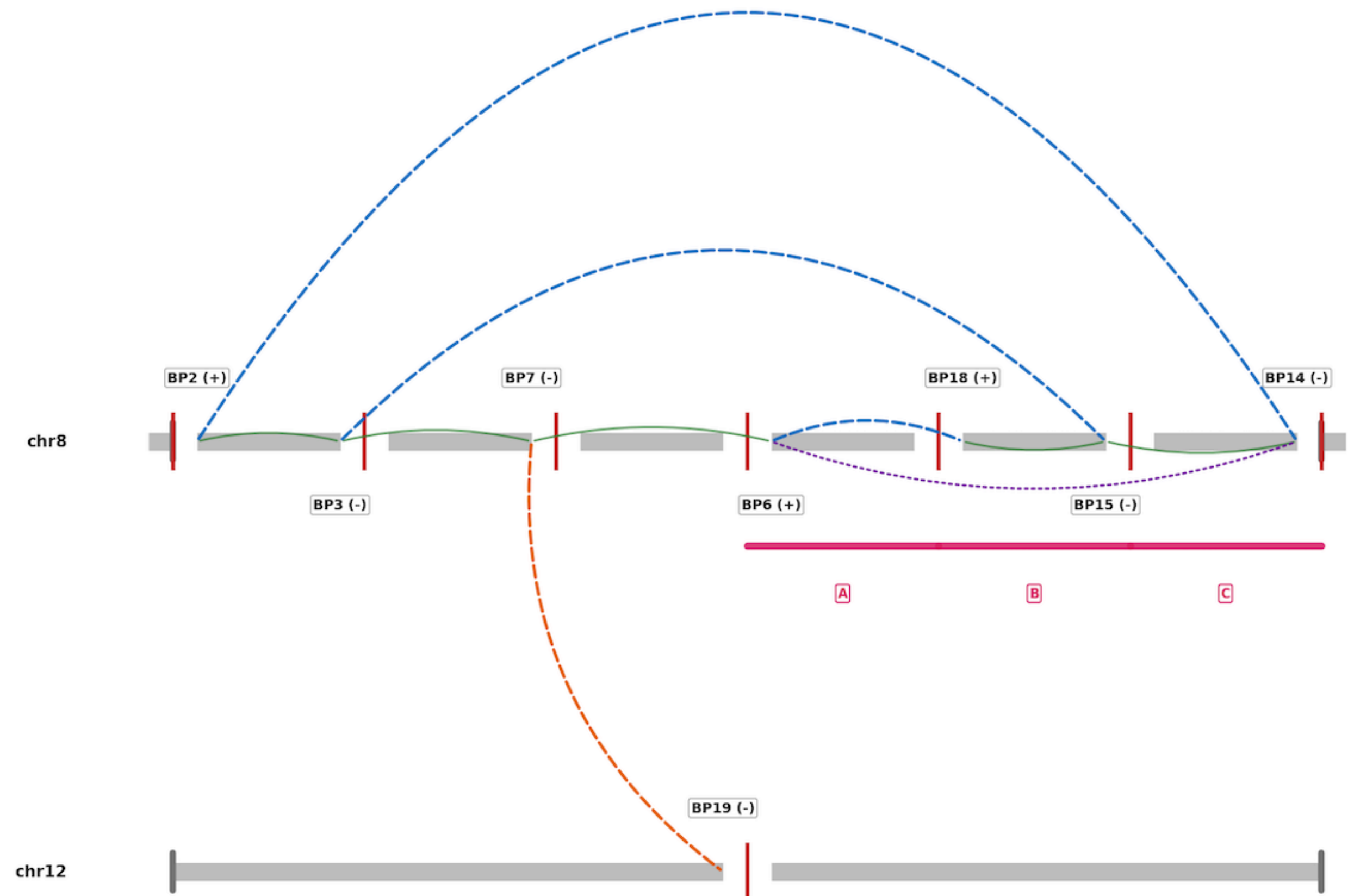
What would a **Deletion Bridge** mean in **P05-1657**

According to my assumption, it looks like **everything** between **BP6** and **BP14** will be **deleted**.

But, in that case, what happens to the breakpoints in between them?

To answer this, I thought why not look at the “**Post Rearrangement Structure**” after all the given rearrangements have taken place.

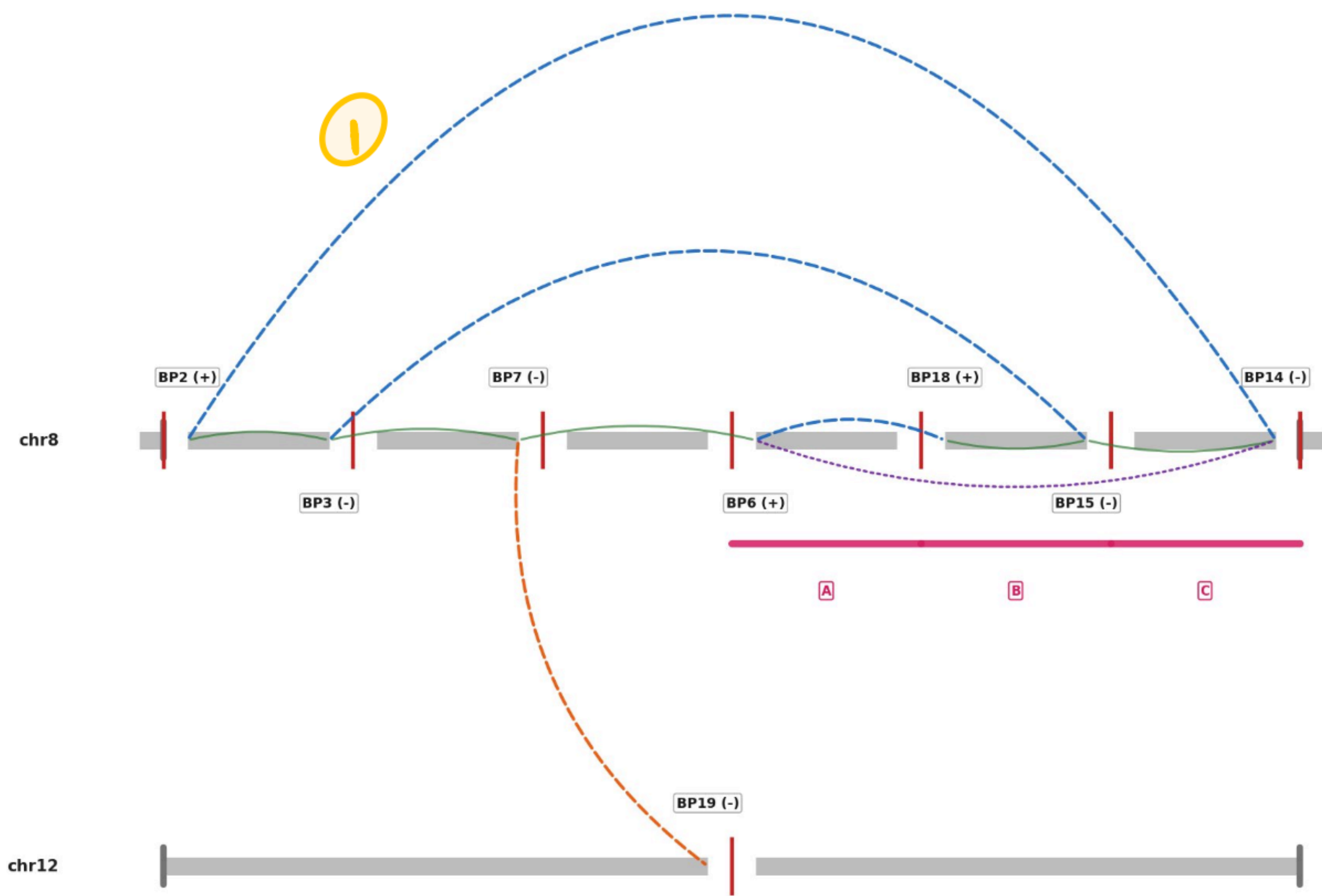
Let’s walk through the rearrangements together.



Rearrangement Walkthrough
For Patient P05-1657

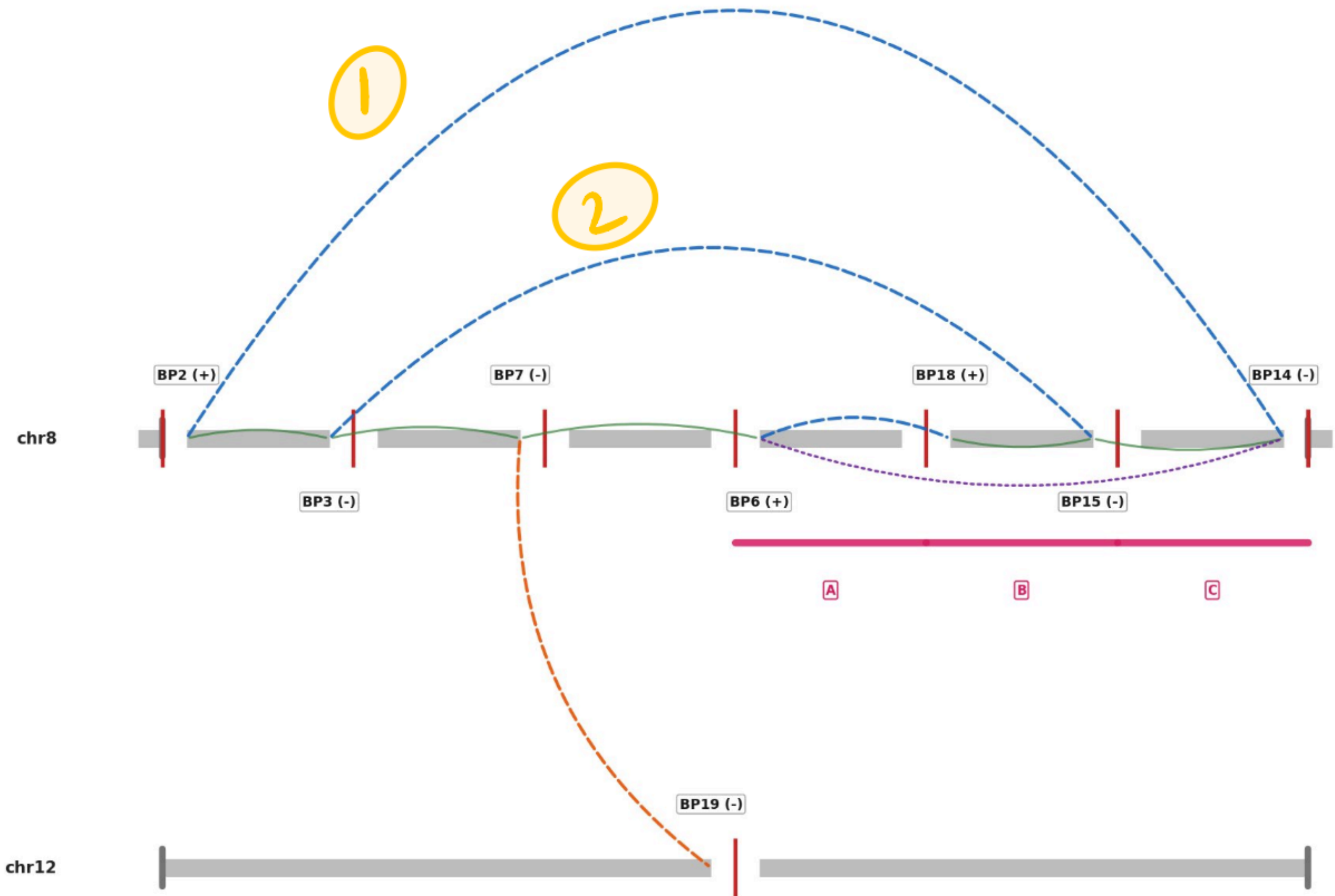
Rearrangement 1

Individual	Chain number	Breakpoint number	Rearrangeme nt number	Chromosome: position	Strand	Supporting reads	Potential chromosomal end loss (left)	Potential chromosomal end loss	Deletion bridge partner breakpoint	Adjacent breakpoint(s)
P05-1657	1	2	11	chr8:40925006	Forward	12	0	0		3
P05-1657	1	14	11	chr8:41035641	Reverse	12	0	0	6	15



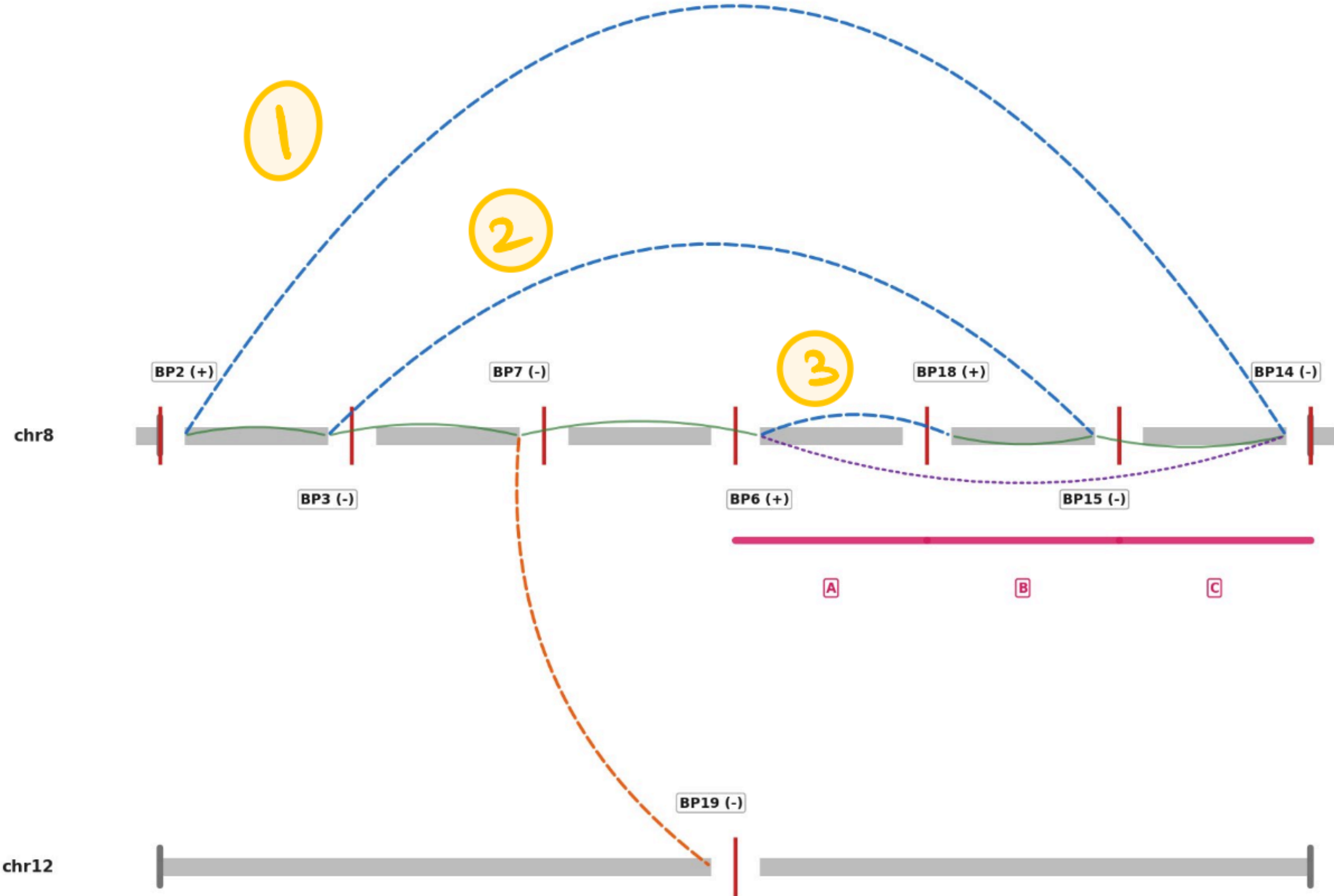
Rearrangement 2

Individual	Chain number	Breakpoint number	Rearrangeme nt number	Chromosome: position	Strand	Supporting reads	Potential chromosomal end loss (left)	Potential chromosomal end loss	Deletion bridge partner breakpoint	Adjacent breakpoint(s)
P05-1657	1	3	17	chr8:40928279	Reverse	10	0	0		2 7
P05-1657	1	15	17	chr8:41032068	Reverse	10	0	0		14 18



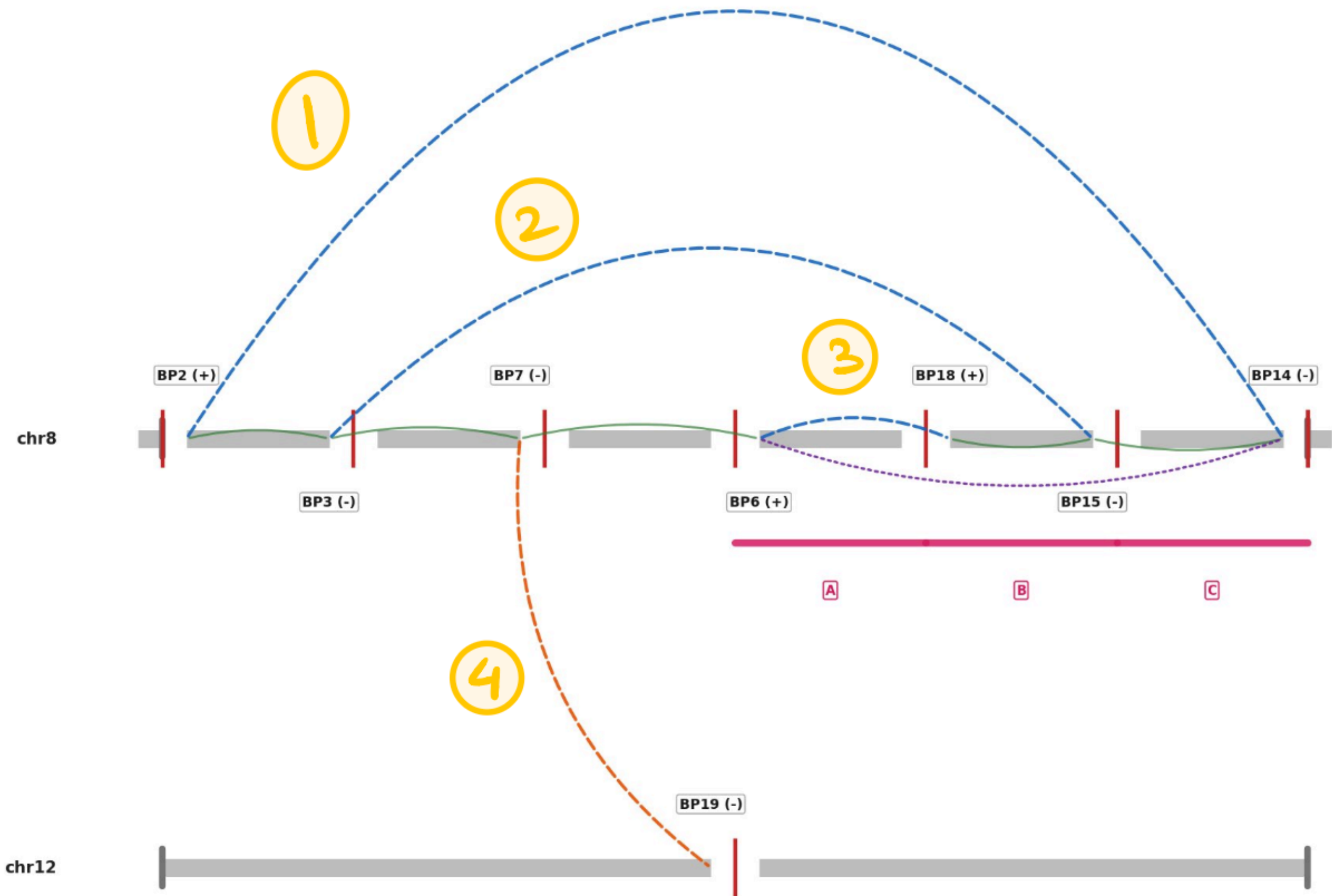
Rearrangement 3

Individual	Chain number	Breakpoint number	Rearrangeme nt number	Chromosome: position	Strand	Supporting reads	Potential chromosomal end loss (left)	Potential chromosomal end loss	Deletion bridge partner breakpoint	Adjacent breakpoint(s)
P05-1657	1	6	165	chr8:40938607	Forward	5	0	0	14	7
P05-1657	1	18	165	chr8:41028309	Forward	5	0	0		15



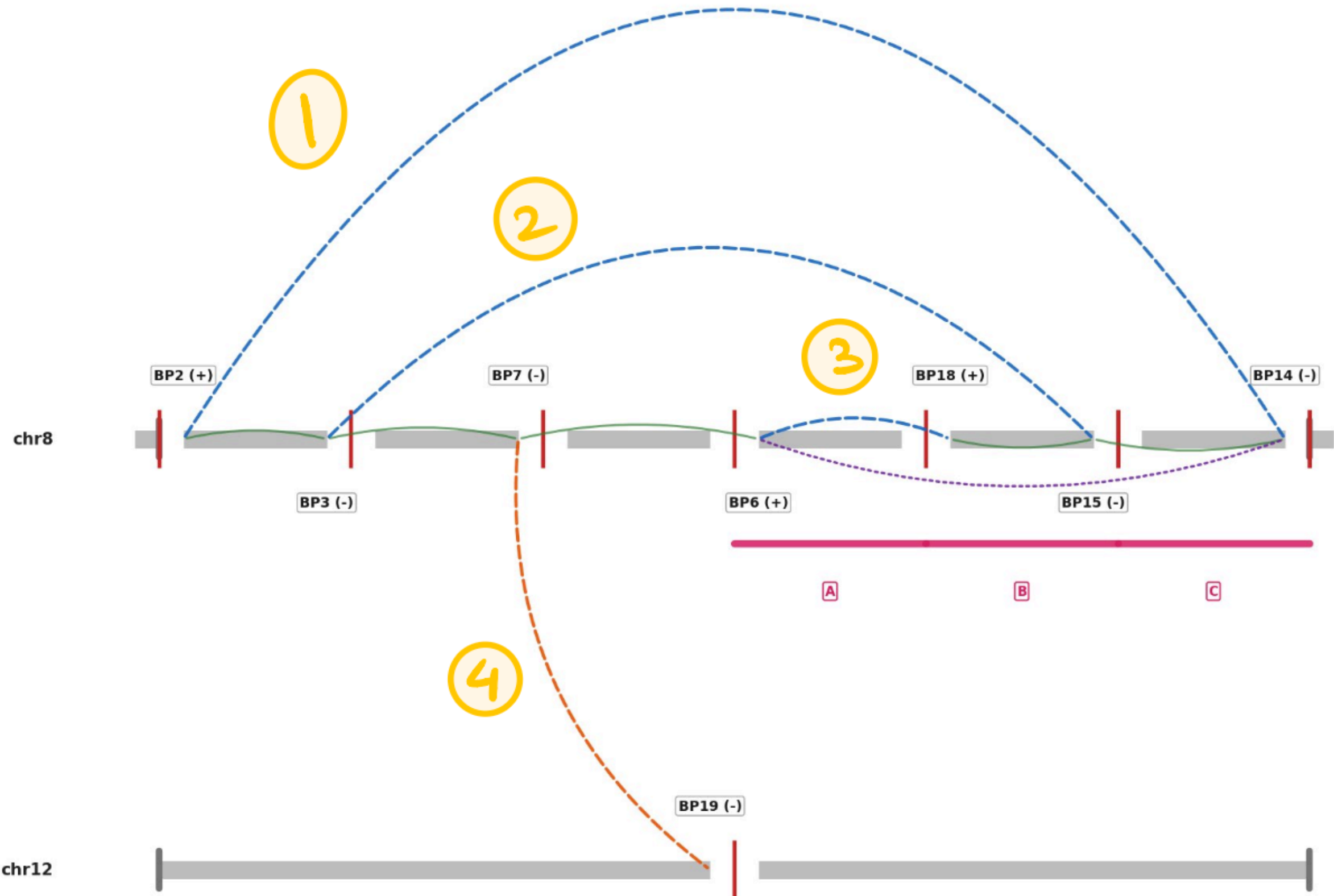
Rearrangement 4

Individual	Chain number	Breakpoint number	Rearrangeme nt number	Chromosome: position	Strand	Supporting reads	Potential chromosomal end loss (left)	Potential chromosomal end loss	Deletion bridge partner breakpoint	Adjacent breakpoint(s)
P05-1657	1	7	168	chr8:40932538	Reverse	5	1	0		3 6
P05-1657	1	19	168	chr12:28921594	Reverse	5	1	0		



Final Chain with Numbered Rearrangements

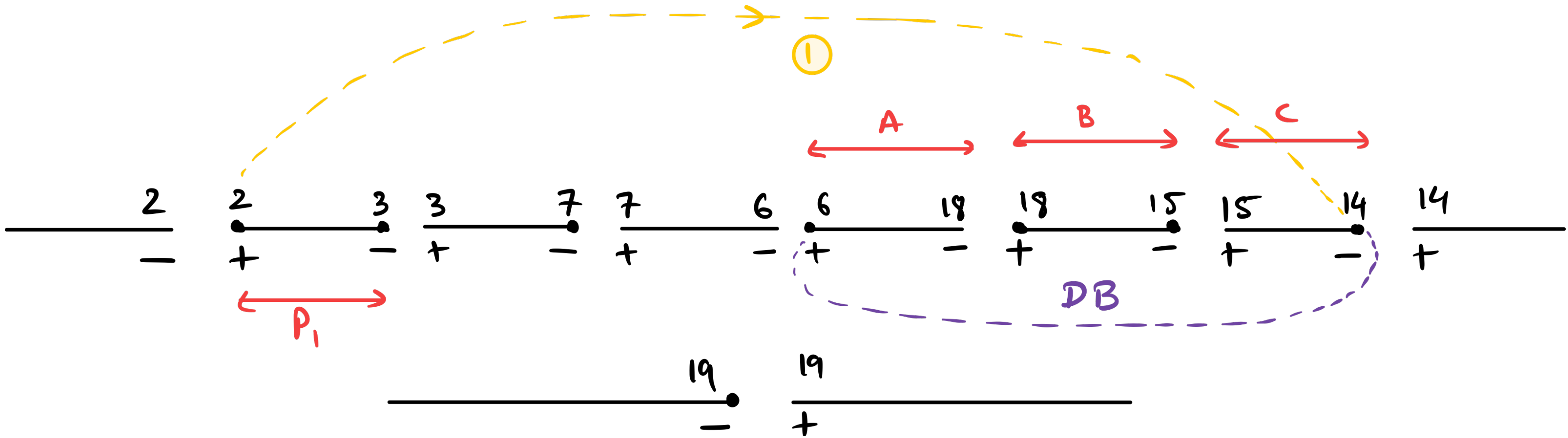
Individual	Chain number	Breakpoint number	Rearrangement number	Chromosome:position	Strand	Deletion bridge partner breakpoint	Adjacent breakpoint(s)
P05-1657	1	2	11	chr8:40925006	Forward		3
P05-1657	1	3	17	chr8:40928279	Reverse		2 7
P05-1657	1	7	168	chr8:40932538	Reverse		3 6
P05-1657	1	6	165	chr8:40938607	Forward	14	7
P05-1657	1	14	11	chr8:41035641	Reverse	6	15
P05-1657	1	15	17	chr8:41032068	Reverse		14 18
P05-1657	1	18	165	chr8:41028309	Forward		15
P05-1657	1	19	168	chr12:28921594	Reverse		



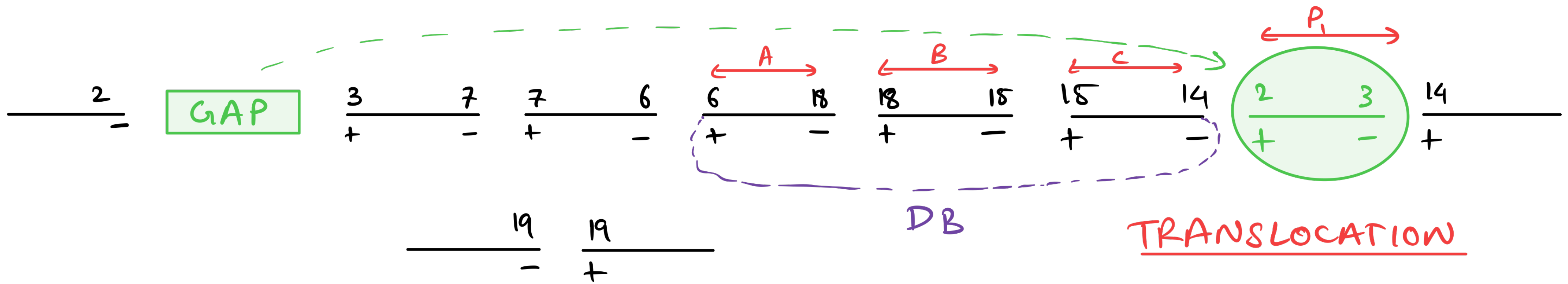
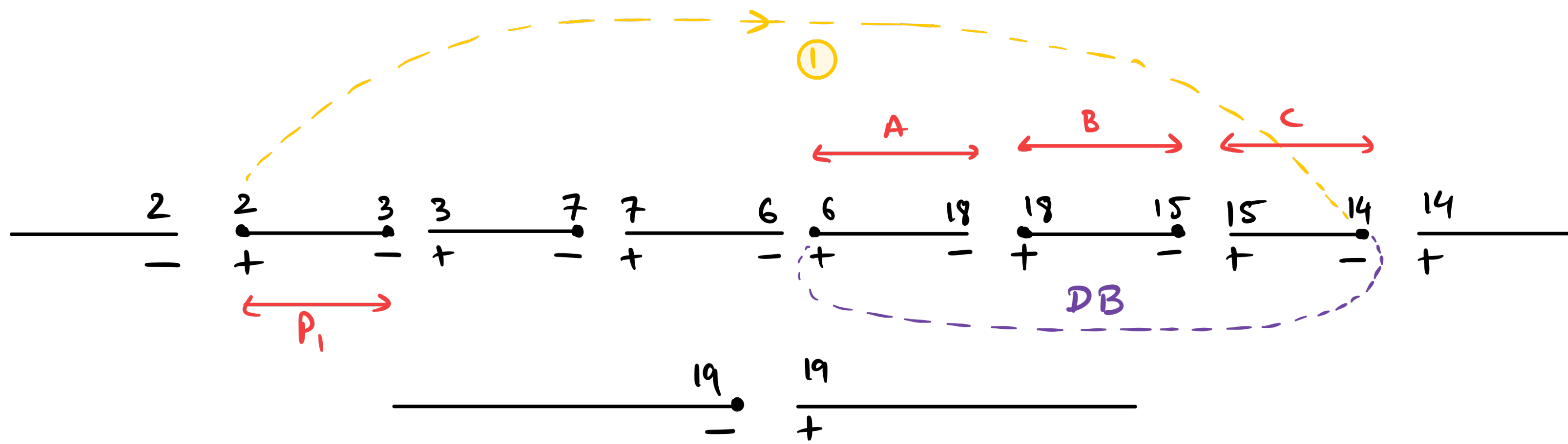
**# Constructing Final Rearranged
Structure For Patient P05-1657**

Rearrangement 1

Individual	Chain number	Breakpoint number	Rearrangeme nt number	Chromosome: position	Strand	Supporting reads	Potential chromosomal end loss (left)	Potential chromosomal end loss	Deletion bridge partner breakpoint	Adjacent breakpoint(s)
P05-1657	1	2	11	chr8:40925006	Forward	12	0	0		3
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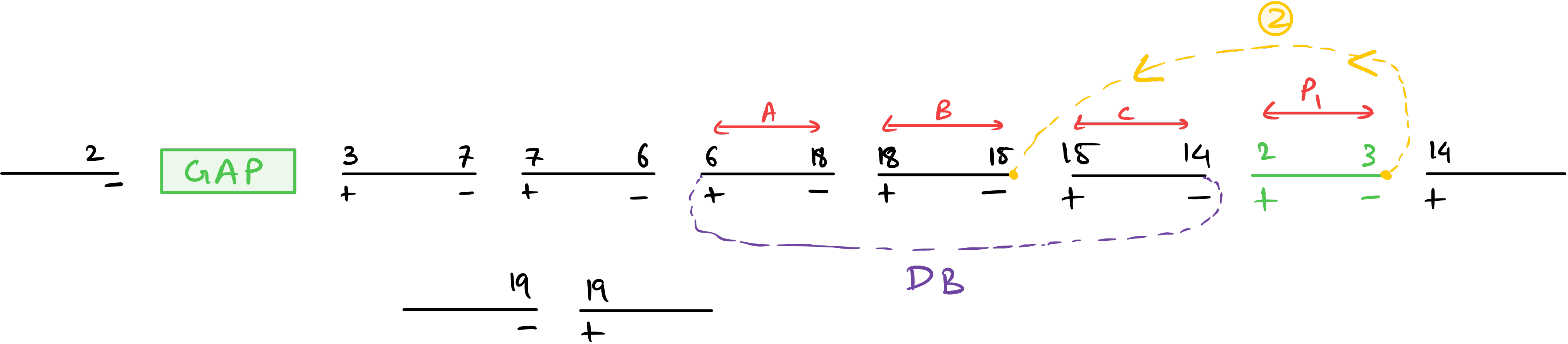


Rearrangement 1

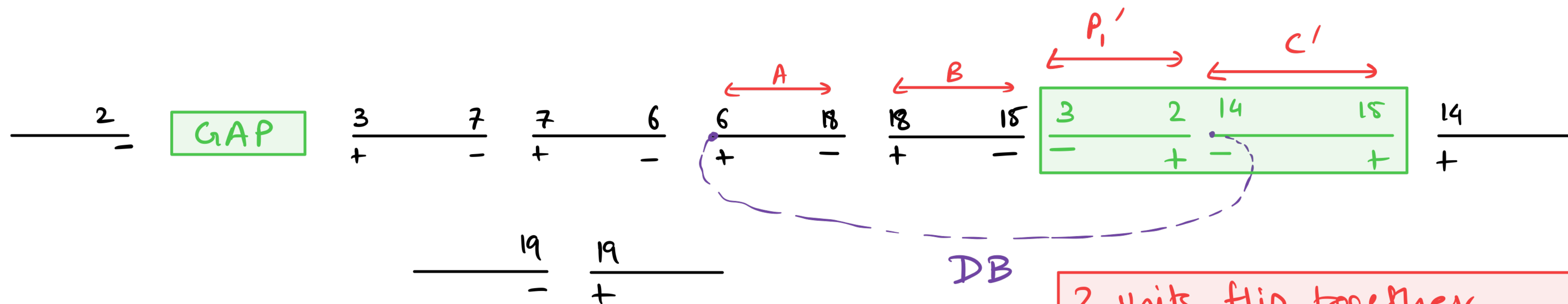
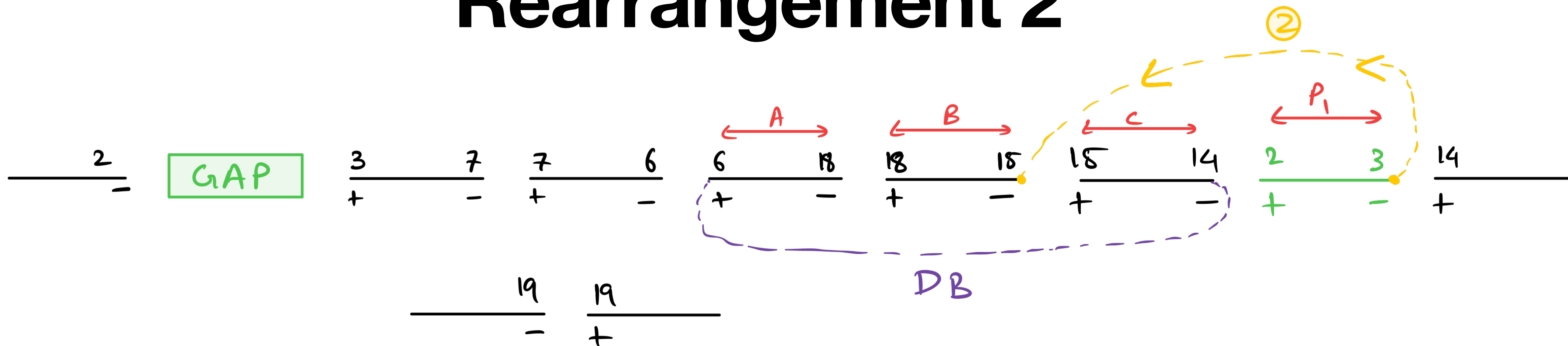


Rearrangement 2

Individual	Chain number	Breakpoint number	Rearrangeme nt number	Chromosome: position	Strand	Supporting reads	Potential chromosomal end loss (left)	Potential chromosomal end loss	Deletion bridge partner breakpoint	Adjacent breakpoint(s)
P05-1657	1	3	17	chr8:40928279	Reverse	10	0	0		2 7
P05-1657	1	15	17	chr8:41032068	Reverse	10	0	0		14 18



Rearrangement 2

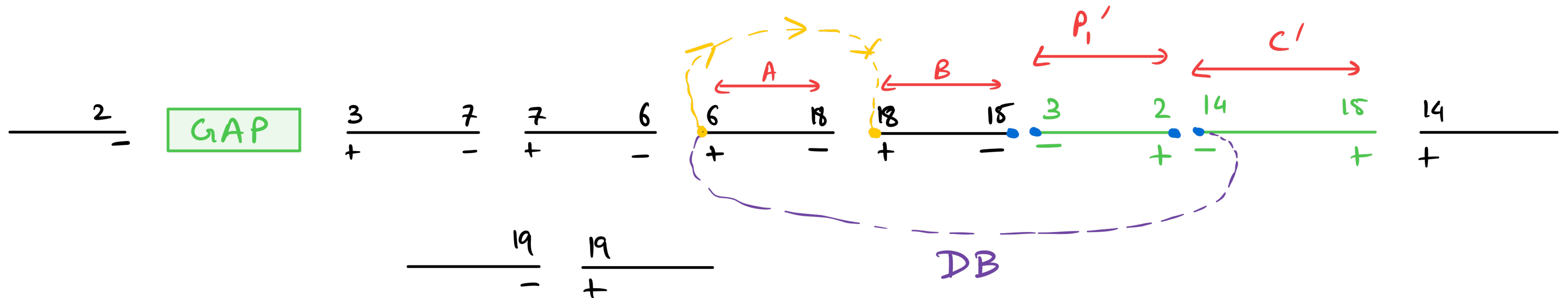


2 units flip together

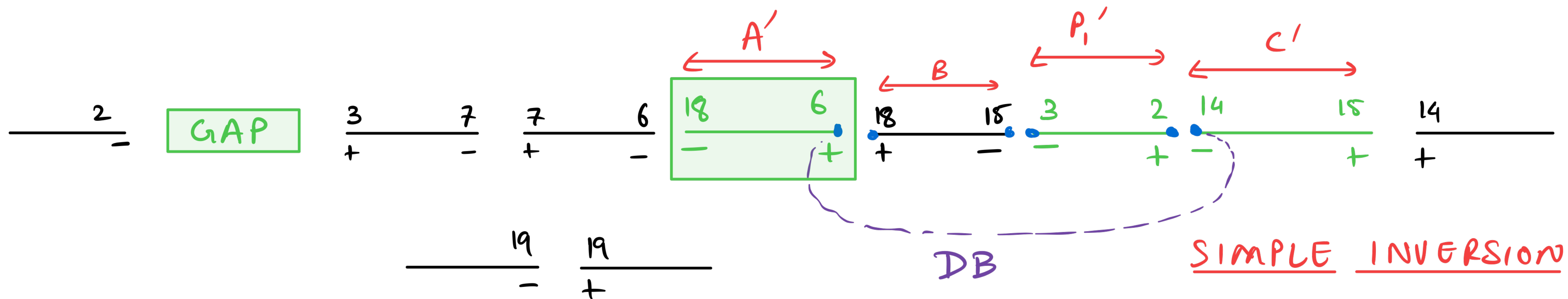
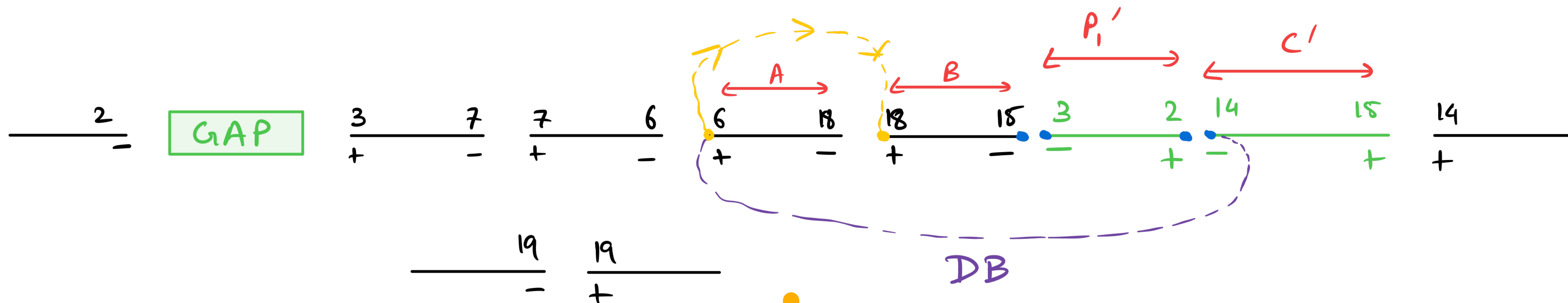
$$(CP_1)' = P_1'c' \rightarrow \underline{\text{UNIT INVERSION}}$$

Rearrangement 3

Individual	Chain number	Breakpoint number	Rearrangement number	Chromosome: position	Strand	Supporting reads	Potential chromosomal end loss (left)	Potential chromosomal end loss	Deletion bridge partner breakpoint	Adjacent breakpoint(s)
P05-1657	1	6	165	chr8:40938607	Forward	5	0	0	14	7
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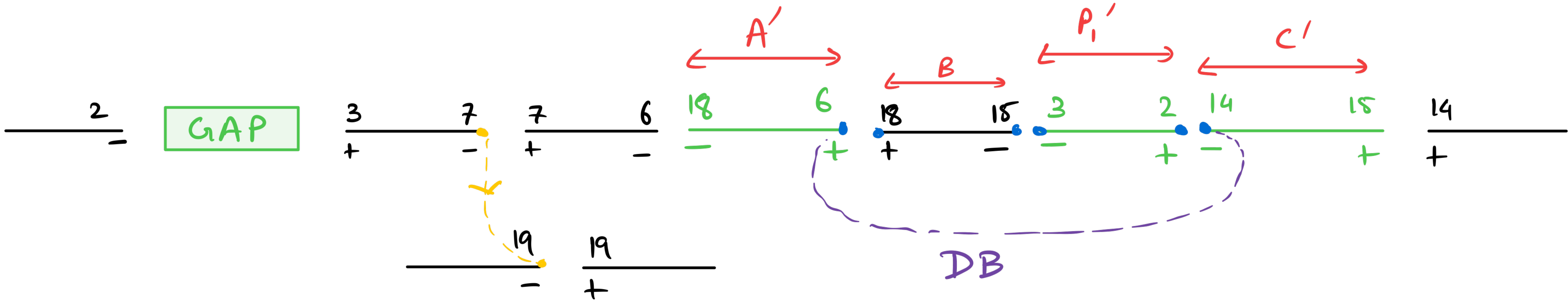


Rearrangement 3

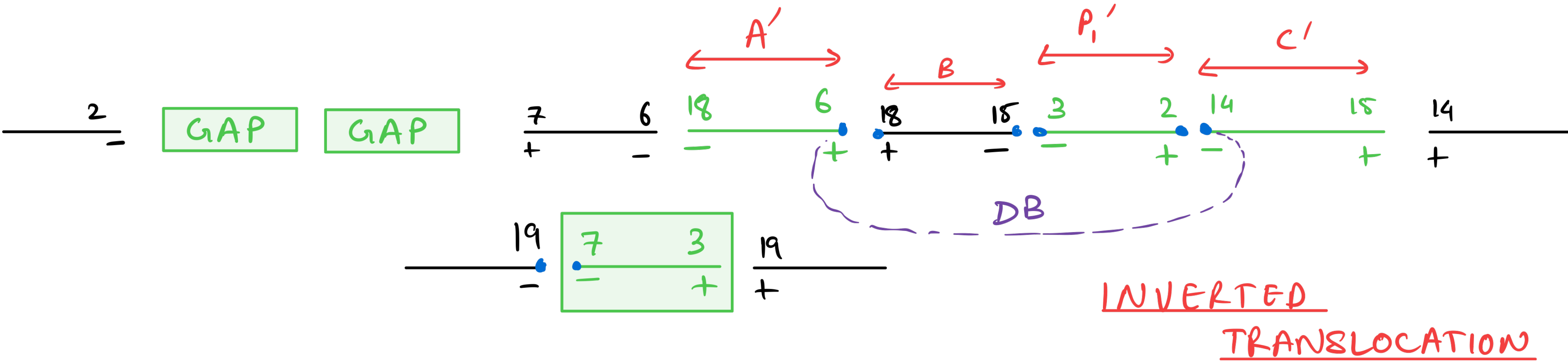
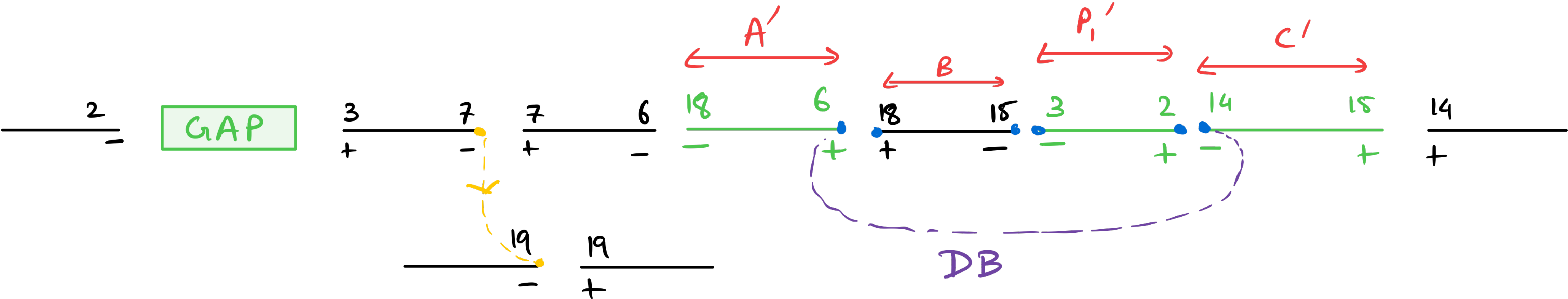


Rearrangement 4

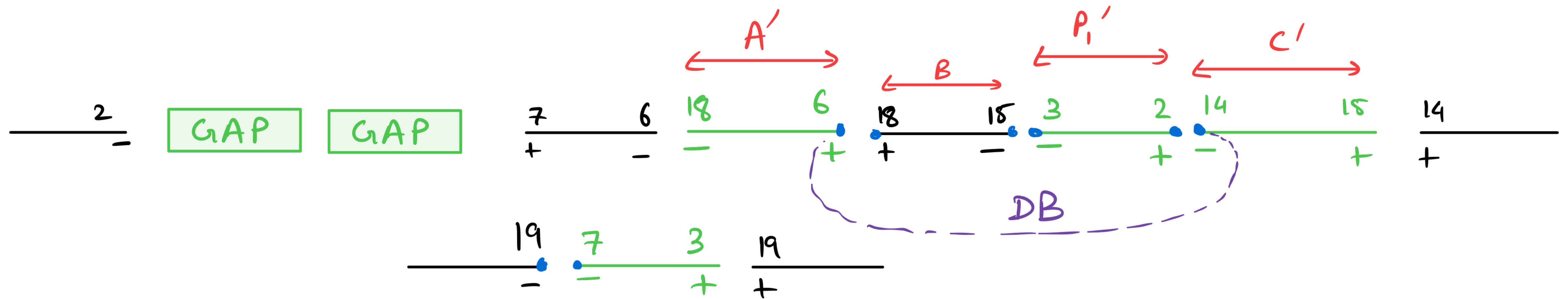
Individual	Chain number	Breakpoint number	Rearrangeme nt number	Chromosome: position	Strand	Supporting reads	Potential chromosomal end loss (left)	Potential chromosomal end loss	Deletion bridge partner breakpoint	Adjacent breakpoint(s)
P05-1657	1	7	168	chr8:40932538	Reverse	5	1	0		3 6
P05-1657	1	19	168	chr12:28921594	Reverse	5	1	0		



Rearrangement 4



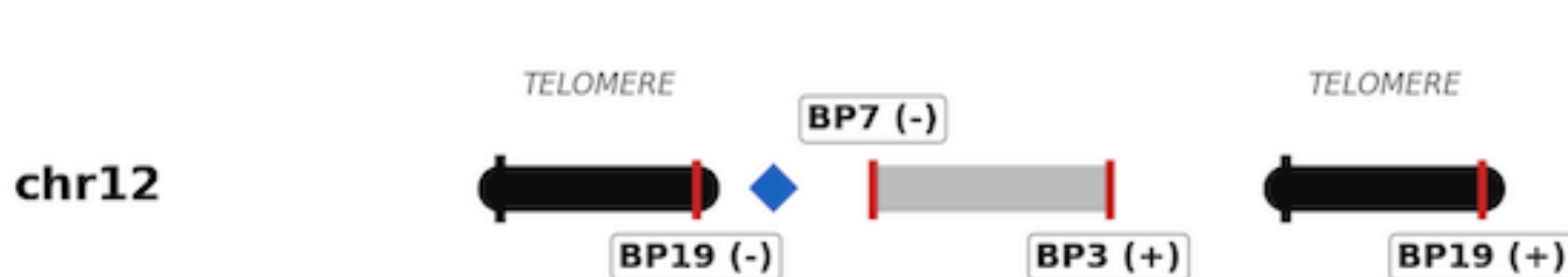
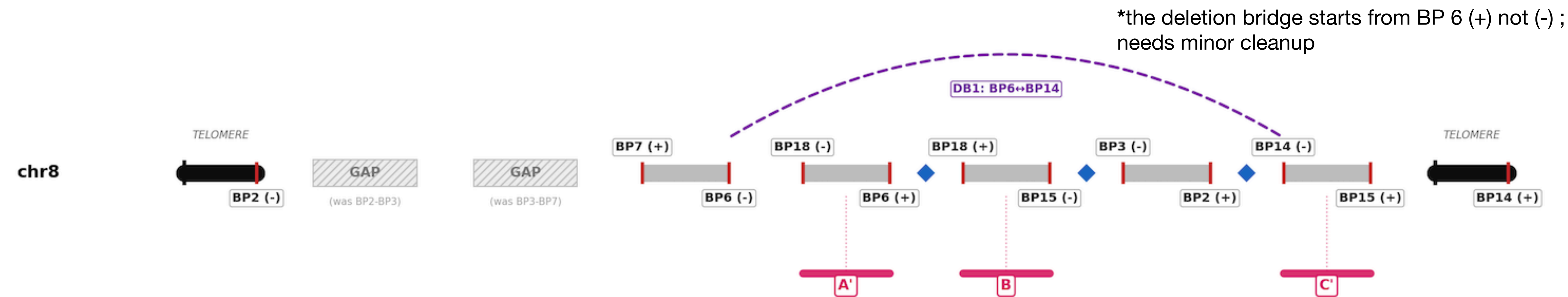
Final Structure After All Rearrangements



Can We Automate This For All Patients?

**Spoiler Alert: Yes, we can!*

Here is the Final Structure for P05-1657 generated using Python



*not actually a telomere end after BP 4 (+) if there **EXISTS** breakpoint data in the **WHOLE** Genome sequence. This only accounts for Baca's chains rows in **mmc5**. The whole genome sequence is in **mmc3** table **S3C**.

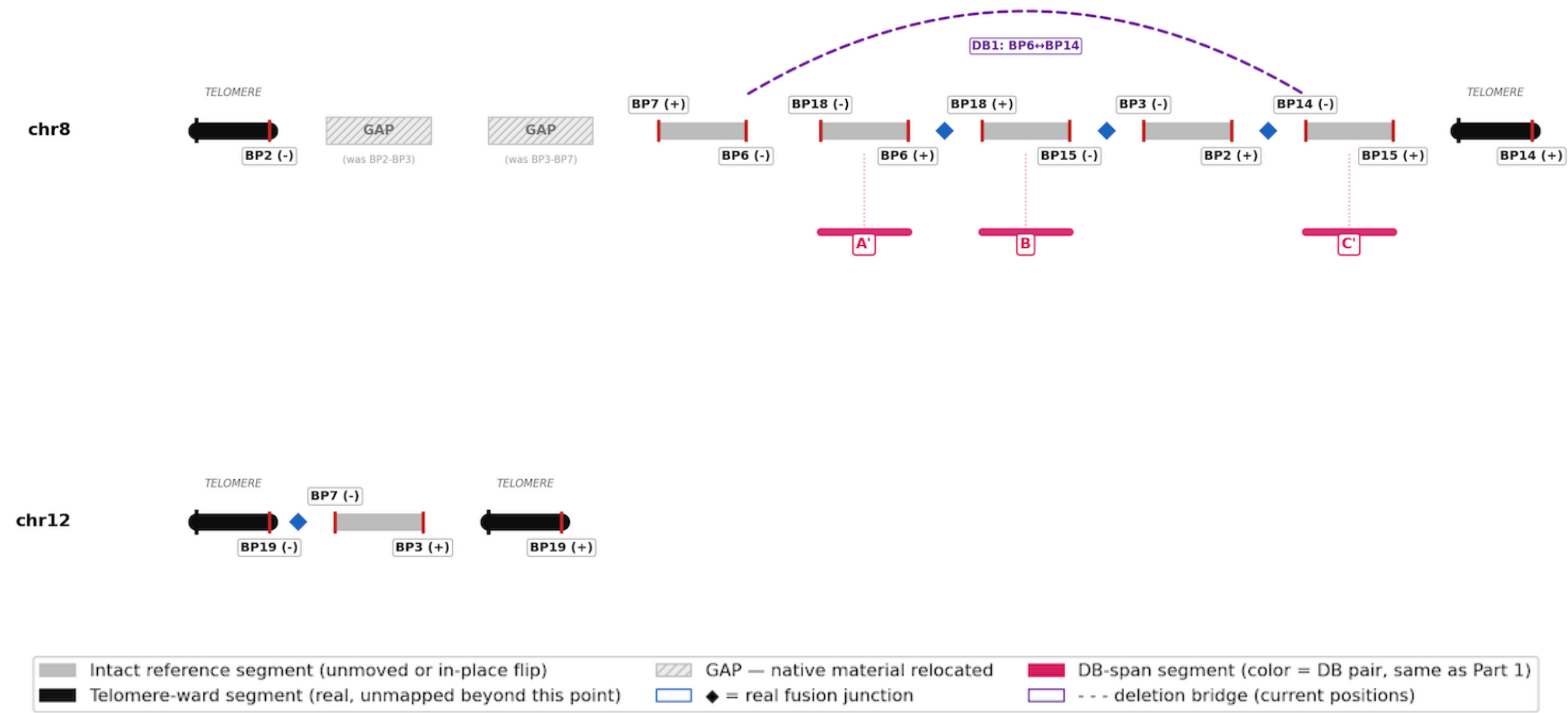
Intact reference segment (unmoved or in-place flip)	GAP — native material relocated	DB-span segment (color = DB pair, same as Part 1)
Telomere-ward segment (real, unmapped beyond this point)	◆ = real fusion junction	- - - deletion bridge (current positions)

There are 2 files for each patient:

First file contains the **structure** itself.

Second file contains the **details** regarding the segments in case the drawing is too compact that it cannot be interpreted.

These 2 files act as a set pair, which exists for each patient, for the final rearranged structure.

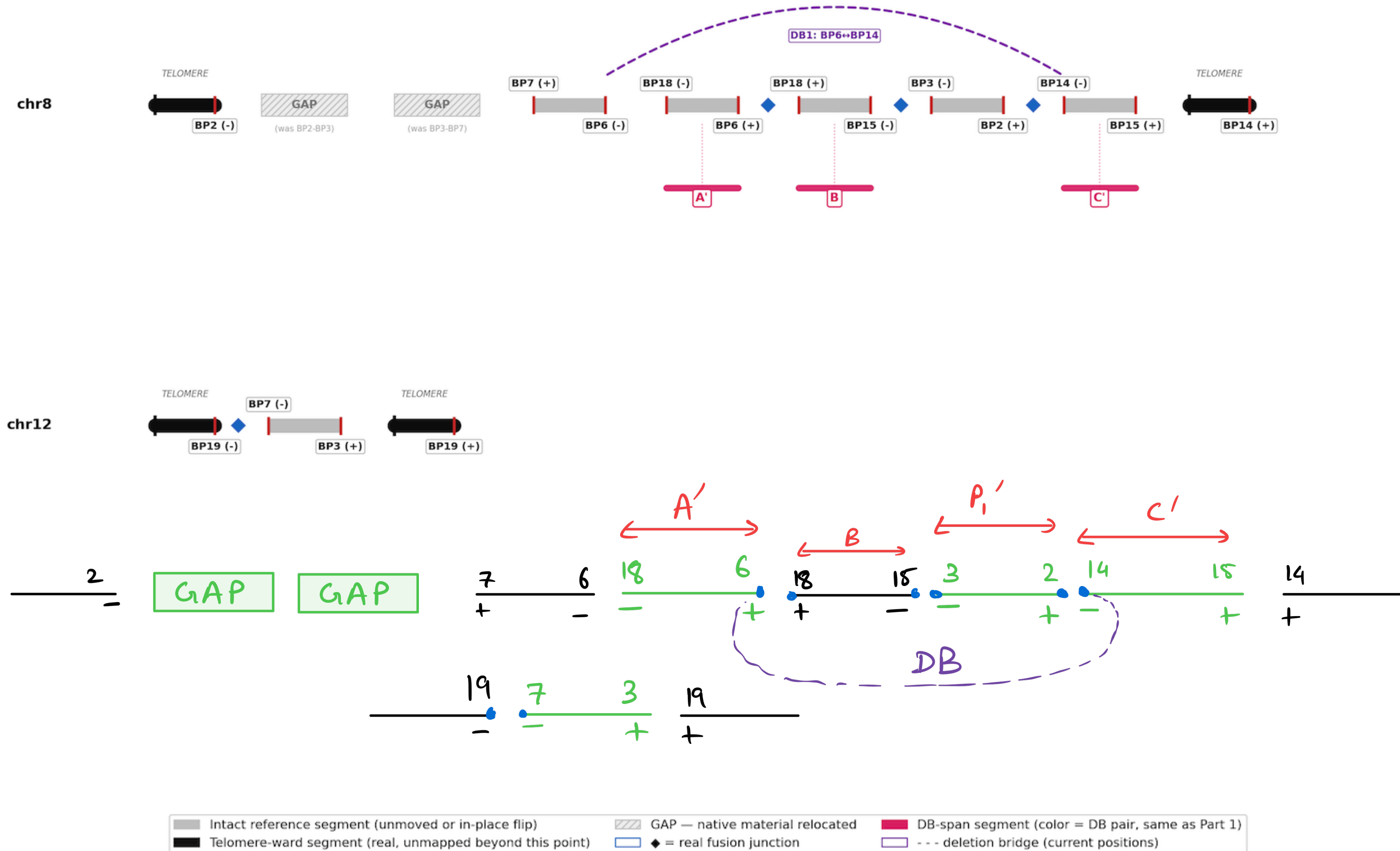


P05-1657 — Baca Chain 1 — FINAL STRUCTURE DETAILS (companion tables for the _final_rearranged_structure image)	
Segment Spans (native) (n=3)	
A	spans 6[+]----18[-]
B	spans 18[+]----15[-]
C	spans 15[+]----14[-]

BP # → Location (n=8)	
BP2	chr8:40,925,006 (+)
BP3	chr8:40,928,279 (-)
BP6	chr8:40,938,607 (+)
BP7	chr8:40,932,538 (-)
BP14	chr8:41,035,641 (-)
BP15	chr8:41,032,068 (-)
BP18	chr8:41,028,309 (+)
BP19	chr12:28,921,594 (-)

Rearrangement Type (n=4)		
#11	BP2-BP14	SIMPLE / NON-RECIPROCAL TRANSLOCATION
#17	BP3-BP15	SIMPLE INVERSION
#165	BP6-BP18	SIMPLE INVERSION
#168	BP7-BP19	INVERTED TRANSLOCATION

Side By Side Comparison with Hand Drawn Structure



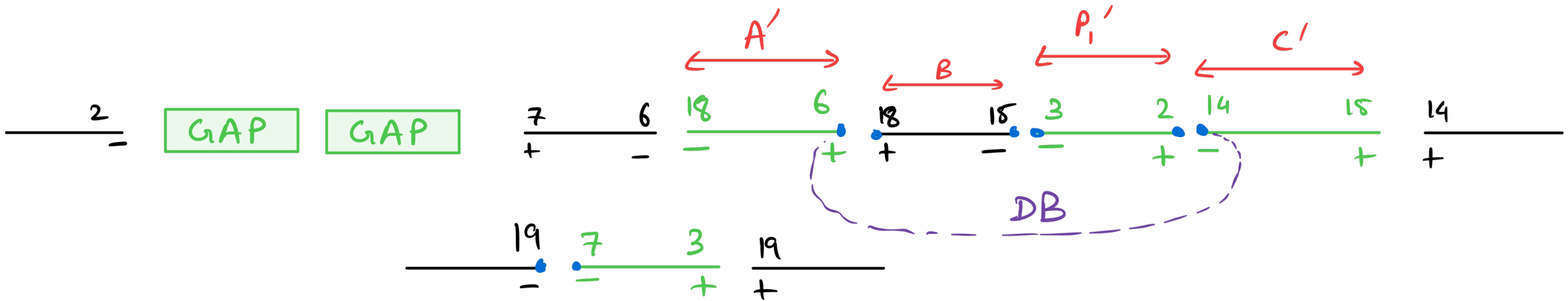
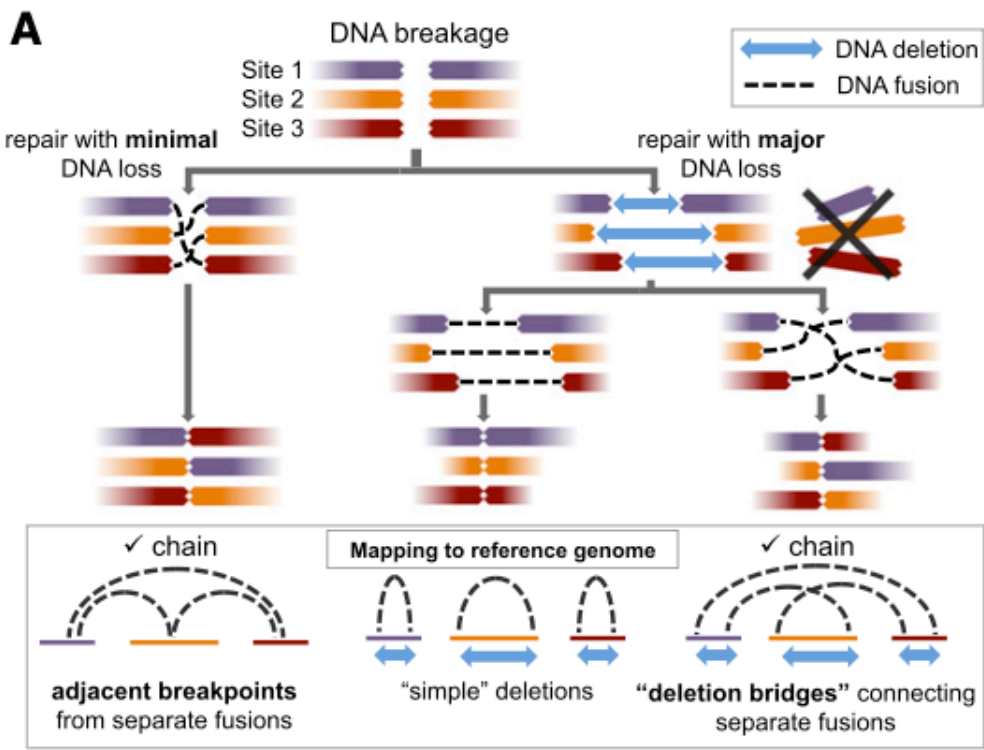
What Baca says a deletion bridge means:

ChainFinder infers a deletion bridge purely from copy number. 2 breakpoints from **separate** fusions (not fused to each other) flanking a segment whose **CN has dropped**.

Baca's own language treats this as **literal loss**:

- Fig. 2A caption: "*The loss of DNA at sites of double-strand breaks may result in simple deletions... or deletion bridges... that span breakpoints from distinct fusions.*"
- Fig. 5B: "*putative TSGs CDKN1B, ETV6, and ETV3 were lost in the context of deletion bridges.*"

So in Baca's framing, a deletion bridge = DNA that left the genome, evidenced by a CN drop between two unrelated fusion breakpoints.



My Conclusion

We rebuilt the real final rearranged molecule for every Deletion Bridge flanked segment in **all 194 mmc5 chains** with a deletion bridge (**521 DB pairs**), using **only Baca's own sequenced fusion edges** and NO invented connections. The flanked material is *essentially* never literally gone:

Verdict	Count	%
DELETED (unaccounted for anywhere)	0/521	0%
Altered – MOVED (relocated elsewhere in the chain)	424/521	81%
Altered – FLIPPED-IN-PLACE (reoriented, not relocated)	57/521	11%
Altered – mixed (both, within one multi-segment pair)	8/521	2%
UNCHANGED	32/521	6%

*this is scoped to the **194 chains** that **actually contain a deletion bridge**, and it only counts as solid because every rearrangement end in those chains resolves to a **real partner within the same chain**.

It is plausible that the flanked segments could be moved outside of Baca's Chain, but if that WERE the case, then Deleted would show a count. But, since it doesn't, we conclude that all flanked segments are **accounted** for in the CURRENT Baca Chains.